

KEYTRUDA[®]

(pembrolizumab)

Defending the Crown: Keytruda's 2028 LOE & Merck's Next-Generation Oncology Strategy

- Keytruda's 2028 LOE creates a significant franchise transition, requiring Merck to simultaneously defend the existing franchise and accelerate the next generation of oncology growth*

DEFEND

ACCELERATE

REPLACE

Keytruda LOE: Defend the Franchise, accelerate the pipeline

1 | Revenue Exposure

Keytruda contributes 57 % of Merck Pharma revenue: estimated \$ 96.6 Billion (2015 – 2025) could be exposed to post – LOE erosion

02 | Clinical Response

Relevant pembrolizumab programs create opportunities across multiple disease areas with key clinical milestones extending through 2028.

03 | Portfolio transition

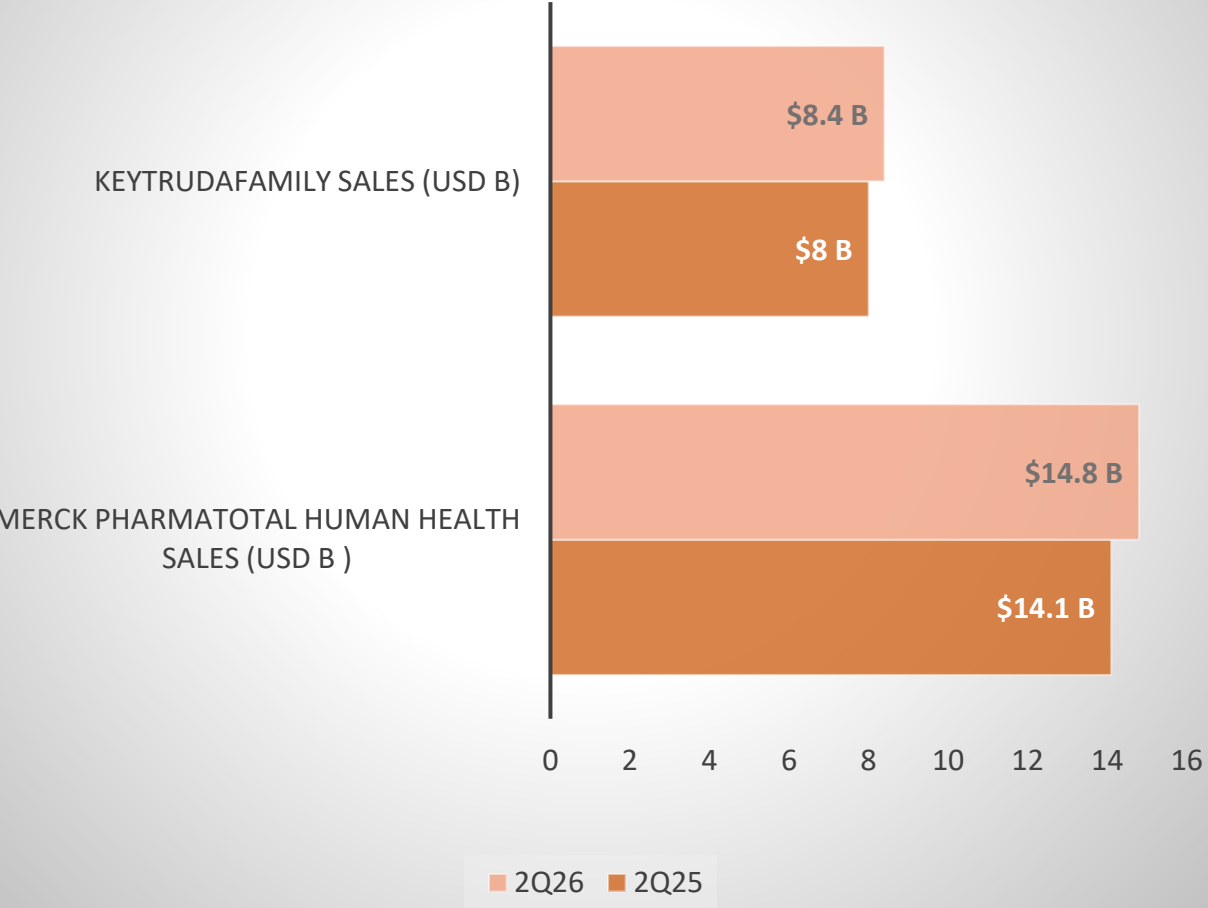
Next – generation oncology assets could provide replacement growth: timing and probability remain the critical constraints.

Strategic question

Can Merck offset Keytruda post- 2028 erosion by defending the existing franchise while accelerating the next generation of oncology growth ?

Keytruda is a portfolio-level concentration for Merck

Keytruda represents ~57% of Merck Human Health sales with both Franchise growing ~5% YoY



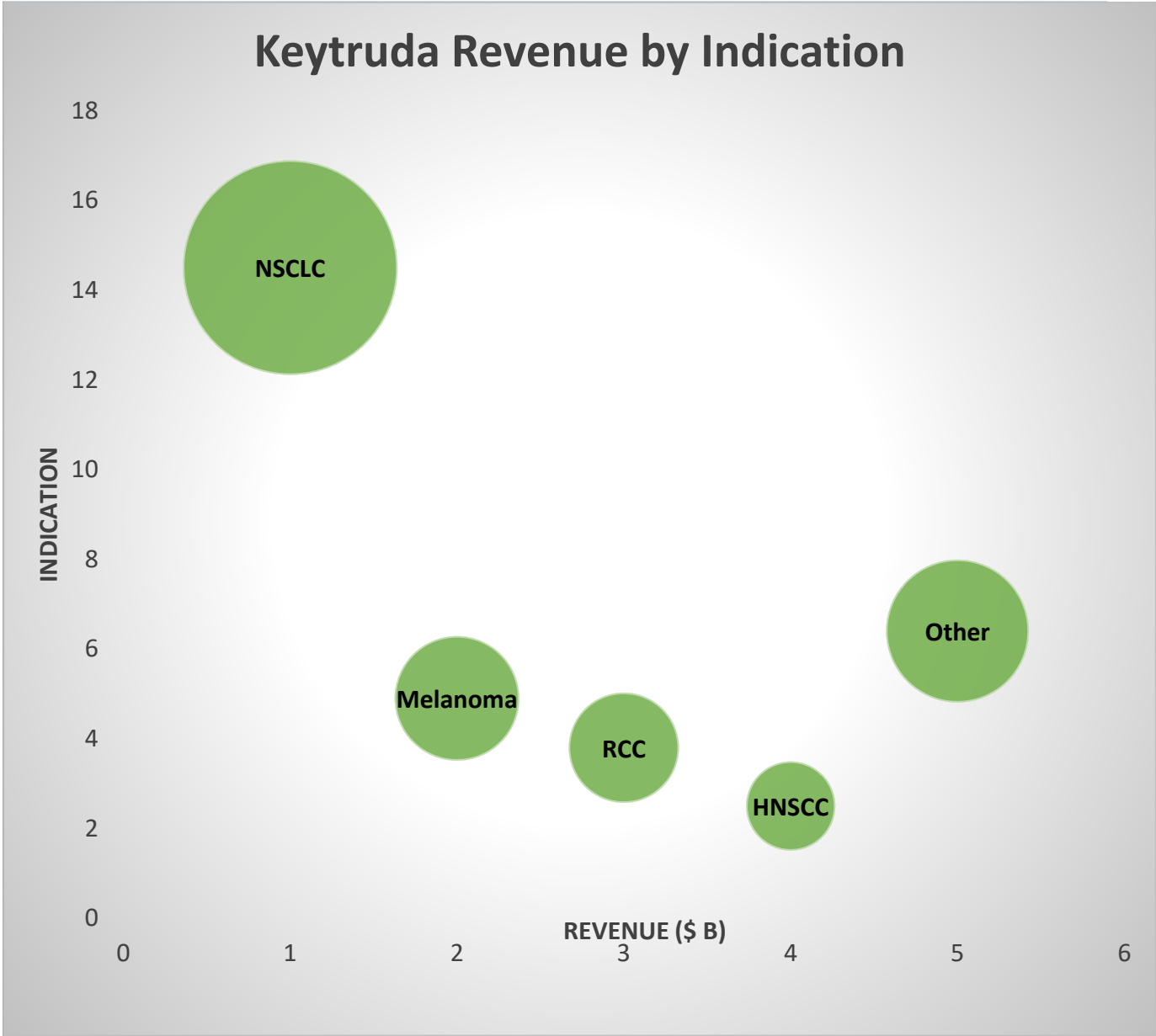
01 | Keytruda Revenue
\$8.4B / FY 2026

02 | YoY growth
+4% / FY 2026

03 | Major geography
North America

04 | Share of Merck’s Pharma Revenue
~56.8%

Keytruda's value is distributed across a broad oncology franchise



01 | Largest opportunity
Indication: NSCLC Market | Revenue: \$ 10.2 Bn
(2024)

02 | Fastest growth
Indication: Triple negative breast cancer
Growth: 5.0 – 5.5 % CAGR (2025 – 2032)

03 | Highest LOE sensitivity
Indication: NSCLC
Rationale: Revenue dependency: 34.6%

Keytruda's protection landscape creates a 2028- 2031 transition window

01 | Core U.S Patent

Primary composition –of-matter patent expires Dec 2028, opens door to biosimilar competition

02 | Additional protection

Two composition patents extend to May/Nov 2029 → 1 – year litigation buffer before biosimilar entry

03 | Indication-specific exclusivity

Some orphan indication extend beyond 2028, additional runway for selected uses.



2026

Current Year



2027

Key Phase 3
Readouts



2028

Core Patent
expires



2029

Biosimilar entry



2030

Orphan drug
exclusivity ends



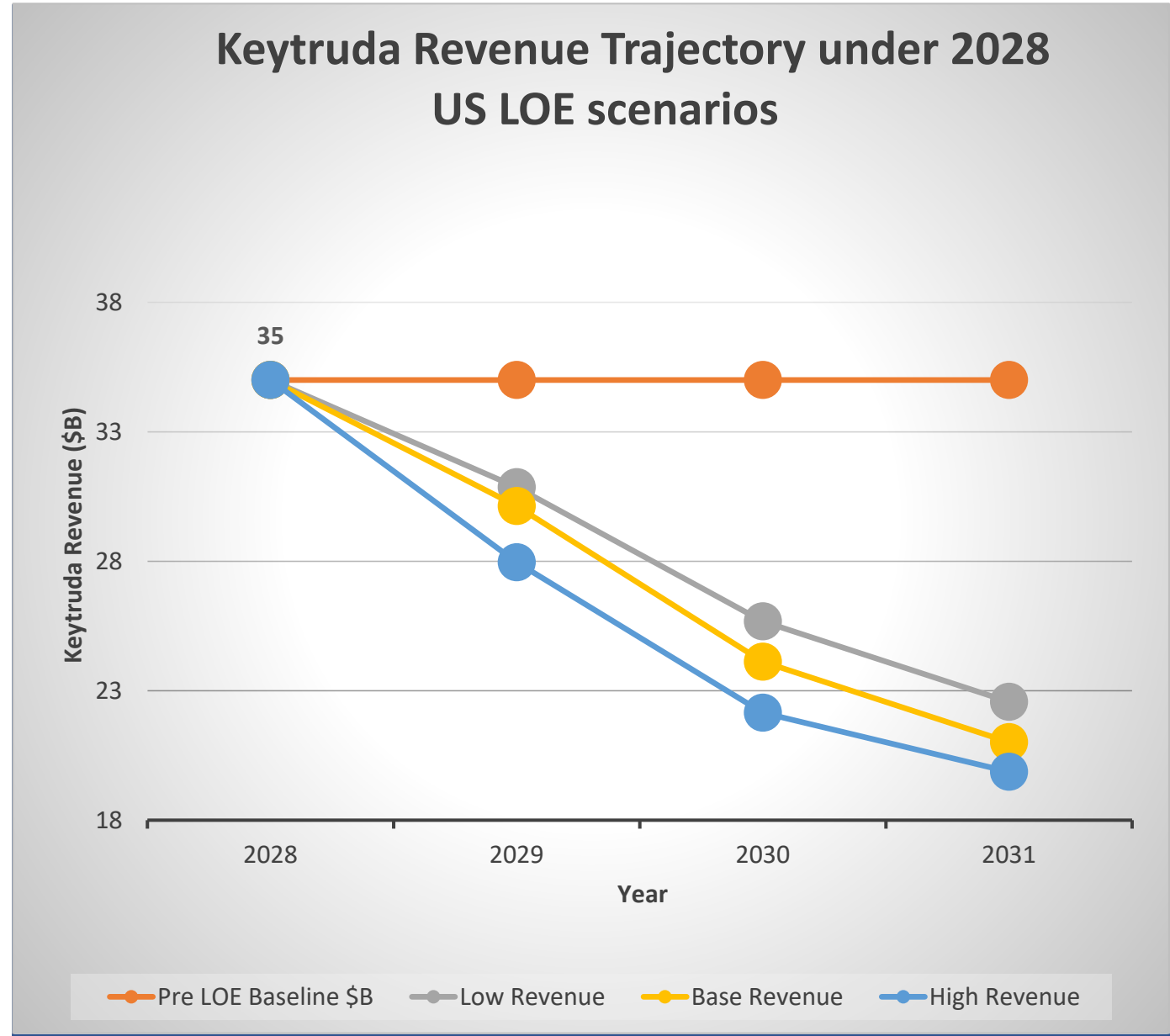
2031

Potential
extension

Clinical Milestone

- ❑ Endometrial Cancer (KEYNOTE-C93) - expected completion May 2027
- ❑ Pediatric Solid Tumors/Lymphoma (KEYNOTE-051) – completion Feb 2028.
- ❑ Esophageal Cancer (MK-3475-06F) – completion June 2028.

The Key economic question is how much Keytruda revenue is exposed after 2028



01 | 2028 Revenue
\$ 35.0 B
(Global Keytruda baseline)

02 | 2030 Revenue at Risk
\$ 10.9 B

03 | Cumulative Exposure
(2029 – 2031)
\$ 29.7 B

Keytruda: Multi-lever franchise defence strategy

01 | Differentiate

Protect the franchise through differentiated delivery

QLEX/ sc administration

- ✓ Faster administration and differentiated patient experience
- ✓ Creates a differentiated formulation alongside IV keytruda

02 | Move earlier

Capture patients before advanced diseases.

Perioperative /adjuvant expansion

- ✓ Move Keytruda from treatment around surgery
- ✓ Increase duration and breadth of franchise exposure

Example: Perioperative bladder cancer

03 | Build combination franchises

Make Keytruda the backbone- not just the molecule

Keytruda + Complementary mechanisms

- ✓ Chemotherapy
- ✓ Targeted therapies / ADCs
- ✓ Novel immunotherapies

Build durable treatment franchises around pembrolizumab

04 | Expand

Increase the addressable patient population

NEW TUMORS

- ✓ Expand into additional cancer types

NEW LINES

- ✓ Earlier / later lines of therapy

NEW POPULATION

- ✓ Biomarker – defined and underserved populations

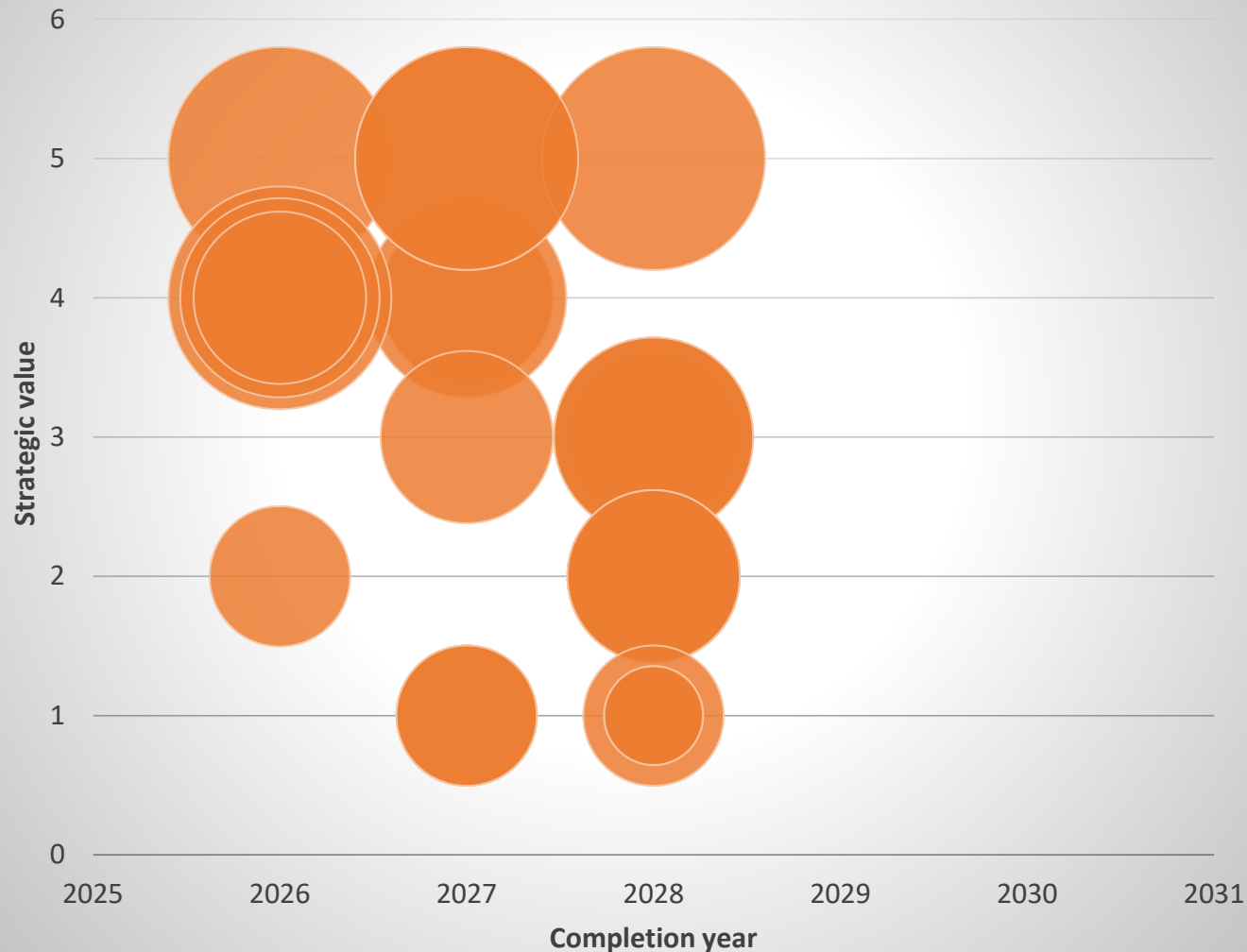
Tumor types -> Treatment lines -> Patient segments

Pembrolizumabs clinical pipeline provides multiple routes to franchise expansion

NCT Number / Trial ID	Indication	Phase	Status	Completion	Strategic Role
NCT05173987	Endometrial Neoplasms	PHASE 3	ACTIVE_NOT_RECRUITING	27-05-2027	EXPAND
NCT03765918	Endometrial Neoplasms	PHASE 3	ACTIVE_NOT_RECRUITING	27-05-2027	MOVE EARLIER
NCT03713593	Carcinoma, Hepatocellular	PHASE 3	COMPLETED	24-09-2024	COMBINE

The post Keytruda strategy depends on the broader Merck oncology pipeline

PEMBROLIZUMAB Pipeline: Timing Vs Strategic VALUE



01 | Phase III/Filing – visible assets

Programs with the maturity and strategic fit to contribute around 2028 LOE window

→ ACCELERATE

02 | Phase II Assets with differentiated opportunity

Programs that create meaningful franchise value beyond the immediate LOE Window.

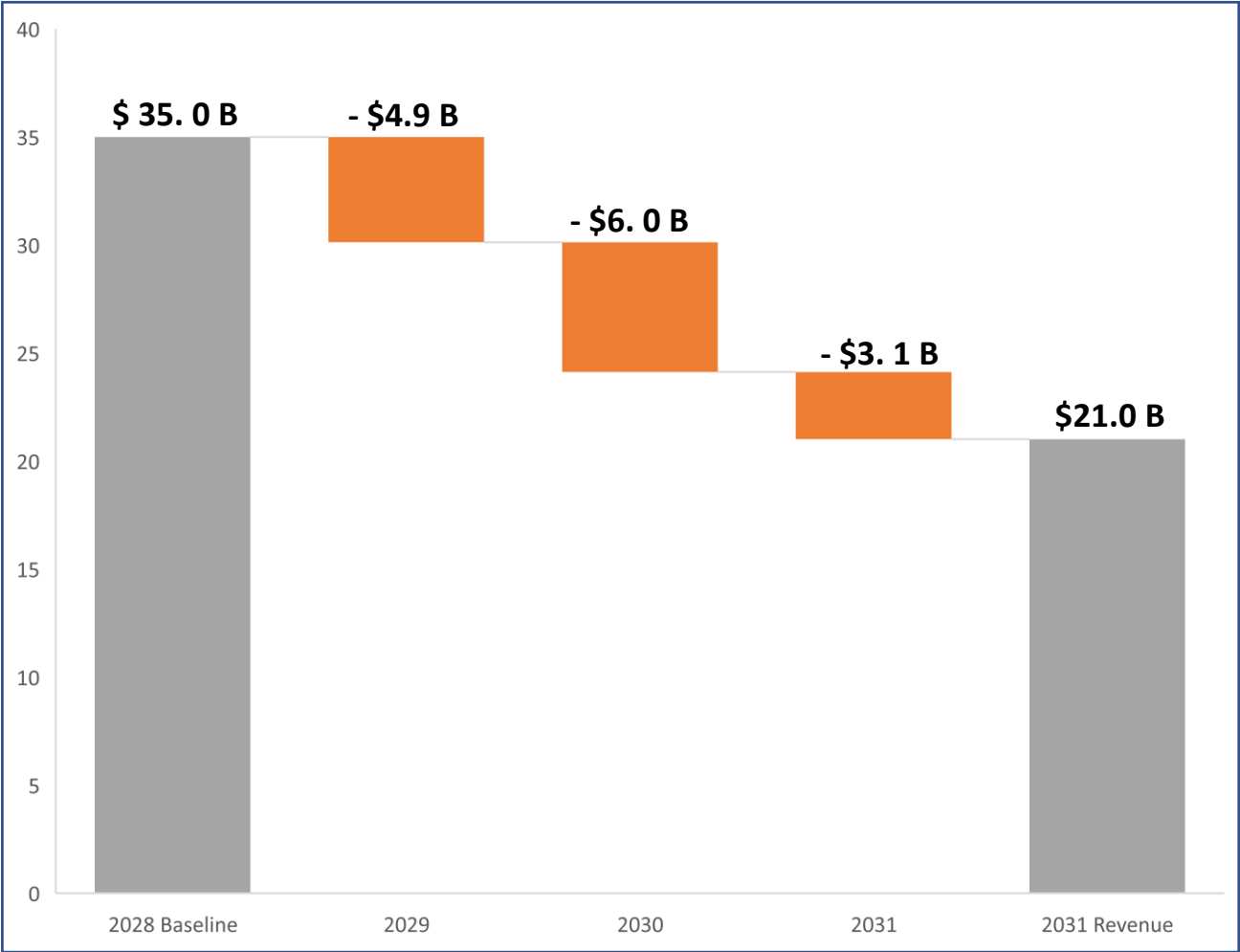
→ BUILD

03 | Early- stage Platforms / mechanisms

Higher-uncertainty programs that could create future oncology growth platforms

→ SELECT / MONITOR

KEYTRUDA's 2028 U.S. LOE could create a \$ 14 B annual revenue Gap by 2031, Requiring multiple replacement Engines



~ \$14 B
Annual gap by 2031

Replacement pathways

01 | Extend

Keytruda Lifecycle

Near-term

02 | Build

New Oncology Growth Engines

Mid- term

03 | Develop

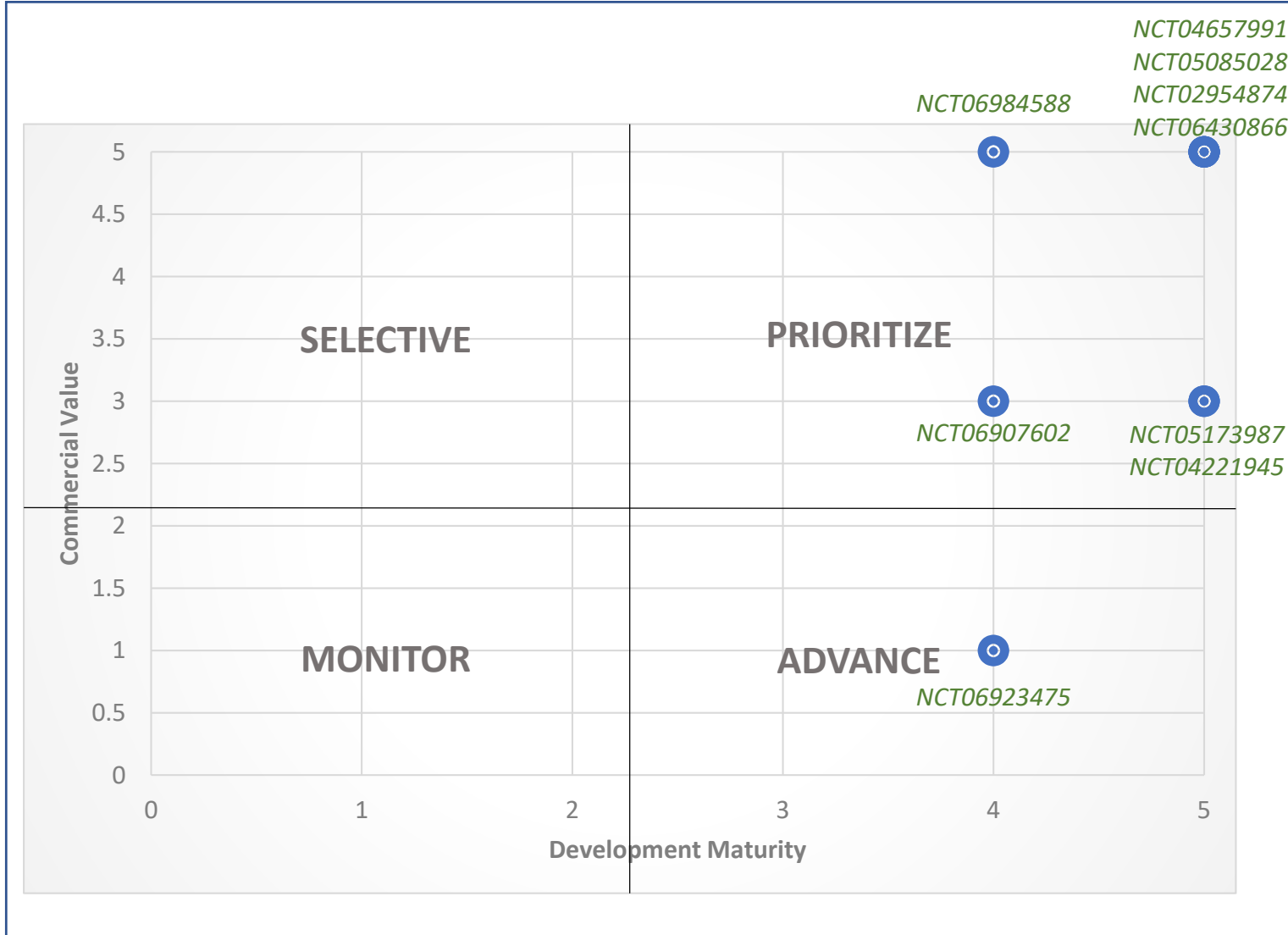
Future Oncology Options

Long - term

\$29.7 B
Cumulative 2029 – 2031 exposure

- Implication: Merck needs a portfolio of replacement engines rather than a one-for-one successor to Keytruda

R&D prioritization should focus on mature high value programs



Prioritise mature programs with meaningful commercial value and near-term relevance to the LOE window

Keytruda's protection landscape creates a 2028- 2031 transition window

01 | Defend

Extend Keytruda runway

2026 – 2028

- ✓ **Differentiate through QLEX**
- ✓ **Expand indications and patient populations**
- ✓ **Move Keytruda earlier in treatment**

Protect near –term franchise value

02 | Accelerate

Convert the pipeline into growth

2027 – 2030

- ✓ **Advance mature, commercially relevant programs**
- ✓ **Build combination- led franchises**
- ✓ **Prioritize opportunities with near term impact**

Bridge the post-LOE revenue gap

03 | Replace

Build the next oncology engines

2029 – 2031+

- ✓ **Develop differentiated mechanisms**
- ✓ **Scale next-generation oncology platforms**
- ✓ **Reduce dependence on a single blockbuster**

Create multiple post-Keytruda growth engines

DEFEND → ACCELERATE → REPLACE

Keytruda's LOE is a portfolio transition – not a single asset replacement challenge

PROTECT THE CORE

Continue lifecycle expansion through differentiated delivery, new indications, earlier use and combinations

CONCENTRATE INVESTMENT

Prioritize mature programs with meaningful commercial opportunity and relevance to the 2028 – 2031 transition.

DIVERSIFY THE FUTURE

Build multiple oncology growth engines across mechanisms and modalities rather than seeking one successor.